GitHub
Tableau

SMARTSTAY ANALYTICS

MULTI-CITY AIRBNB MARKET INTELLIGENCE

Data Preparation
(PYTHON)

→

Data Analysis
(EDA)

→

Insights Visualization
(TABLEAU)

TEAM MEMBERS

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CONTEXT & PROBLEM STATEMENT

CONTEXT

Airbnb holds all market data, but investors, hosts and guests have very limited access to it.
Audience: real-estate investors, hosts, platform managers.

PROBLEM STATEMENT

Where should I invest, how should I price, and what makes a 5-star host?

OBJECTIVE

Replace guesswork with data-driven scoring, KPIs and three Tableau dashboards.

42,904
Listings Analysed

+143%
Superhost Revenue Premium

$255K
Top Neighbourhood Revenue/yr

27.1%
Listings With Zero Revenue

DATA ENGINEERING

DATA SOURCE

Inside Airbnb (insideairbnb.com) — Feb 2026 scrape, 4 cities

ORIGINAL VOLUME

42,904
Total Rows

→

FINAL DATASET

30,048
Cleaned Rows

DATA QUALITY

70% retained | 0 nulls remaining | 52 duplicates removed

3-pass cleaning pipeline

metadata removal → non-business cols → quality fixes

01 Data Collection

Downloaded 42,904 listings from Inside Airbnb across Amsterdam, Antwerp, Athens, Bangkok.

02 Cleaning & EDA

Three-pass pipeline: removed metadata, non-business cols, fixed currency symbols and nulls.

03 Analysis & KPIs

Engineered RevPAR, Investment Potential Score, market segmentation tiers.

04 Visualization

Built three interactive Tableau dashboards for investors, hosts, and platform managers.

KPI & METRICS FRAMEWORK

ACTIVE LISTINGS

30,048

AVG ANNUAL REVENUE

$37,016

SUPERHOST PREMIUM

+143%

AVG REVPAR

$101

ENTIRE HOMES

82.3%

MEDIAN PRICE

$150

How are they computed?

- Derived from cleaned dataset (30,048 listings)
- Calculated using Python (Pandas, NumPy)
- RevPAR = Price × (Occupancy nights / 365); Investment Score = 0.4·Quality + 0.4·Demand + 0.2·log(Revenue)

KEY INSIGHTS (EDA)

Superhosts earn +143% more revenue than regular hosts.

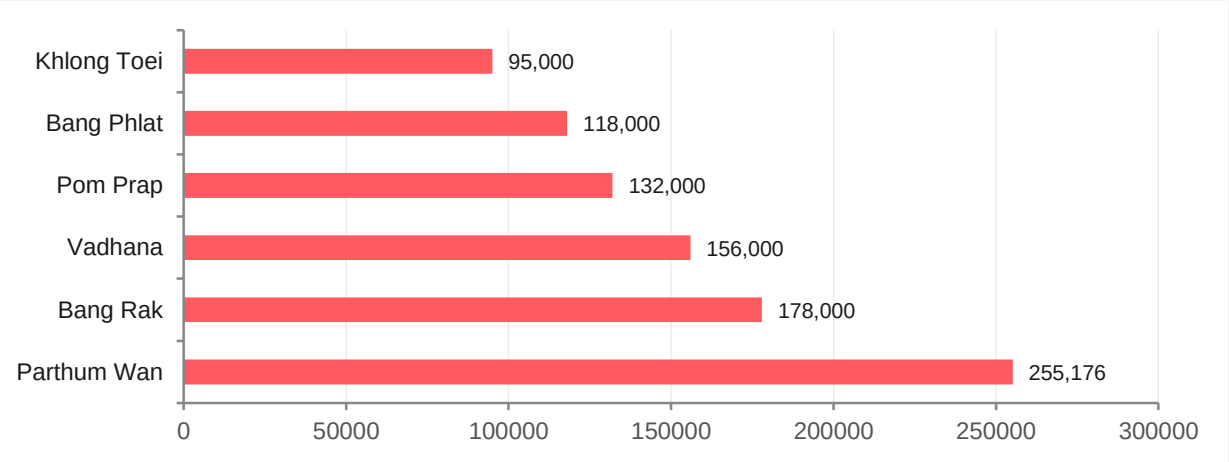
27.1% of listings (8,158) generate zero revenue — major mispricing.

Parthum Wan (Bangkok) tops at $255,176/year average revenue.

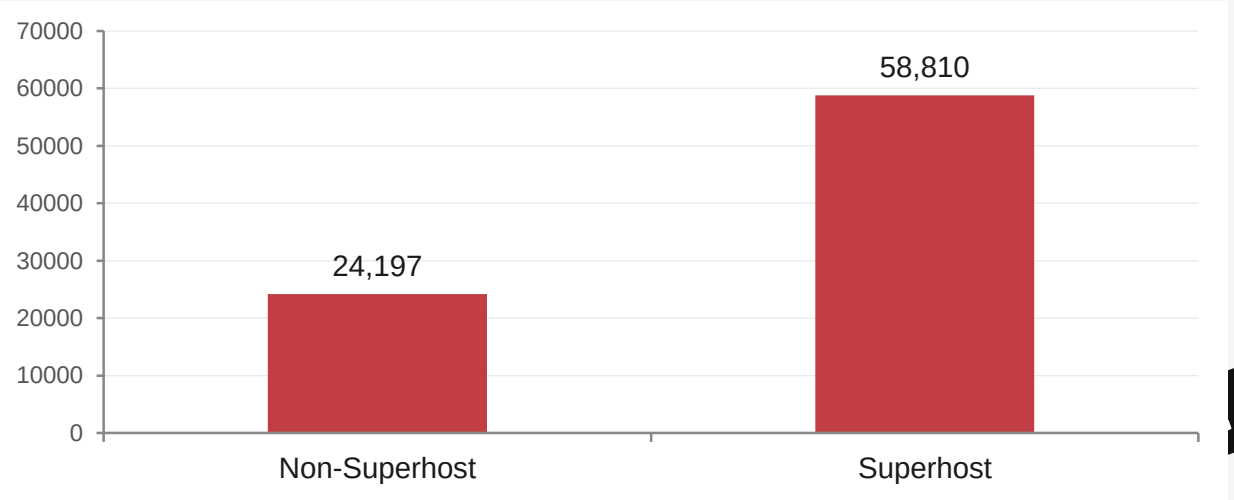
Entire homes dominate with 82.3% market share + highest revenue.

"Value" sub-score (4.705) is the weakest quality dimension.

Top Neighbourhoods — Avg Annual Revenue



Superhost vs Non-Superhost — Avg Revenue



ADVANCED ANALYSIS

ANALYTICAL DEPTH (WHAT WE DID)

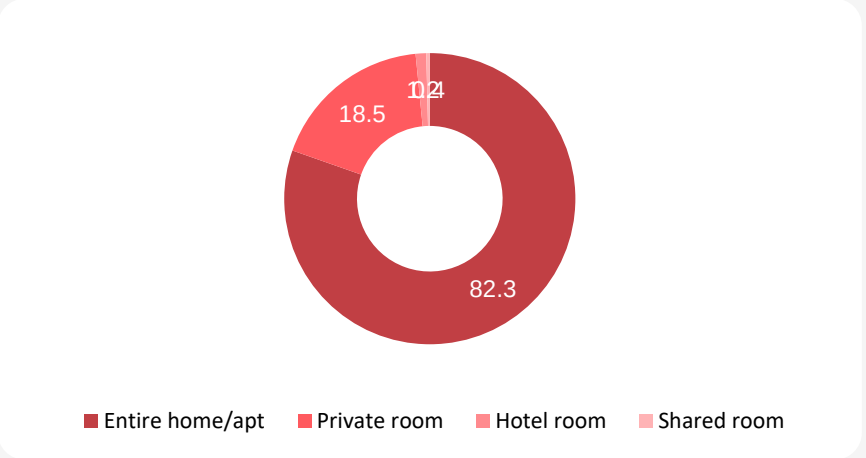
- Segmentation Analysis — by city, neighbourhood, room type and price tier.
- Statistical Testing — Mann-Whitney U for Superhost effect, ANOVA across room types.
- Feature Engineering — built RevPAR and Investment Potential Score (0–100).
- Predictive Modelling — linear regression to predict price from property features.

NEW UNDERSTANDING (BEYOND EDA)

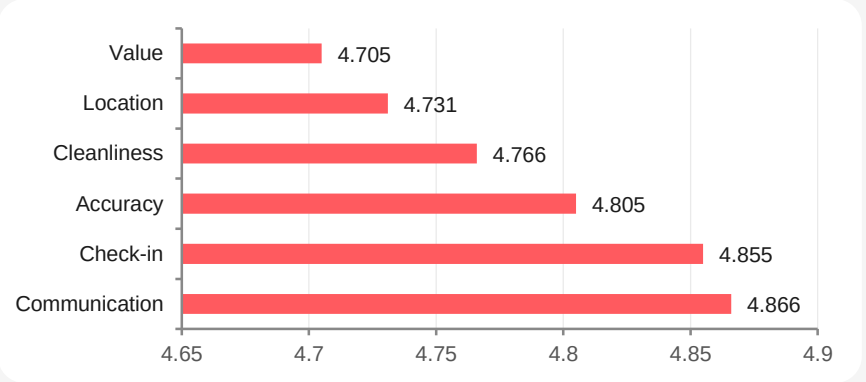
- Mean is misleading — RevPAR mean $101 vs median $12 reveals huge inequality.
- Quality > Price — Superhost status drives revenue more than charging more.
- Mid-tier ($75–$300) is the sweet spot — most consistent returns.
- Hidden gems exist — Schoonbroek-Rozemaai (Antwerp) tops Investment Score at 81.85.
- Instant Book trade-off — +3% occupancy, only −0.07 rating cost.

Python (Pandas, NumPy, SciPy, Scikit-learn)

Market Composition by Room Type

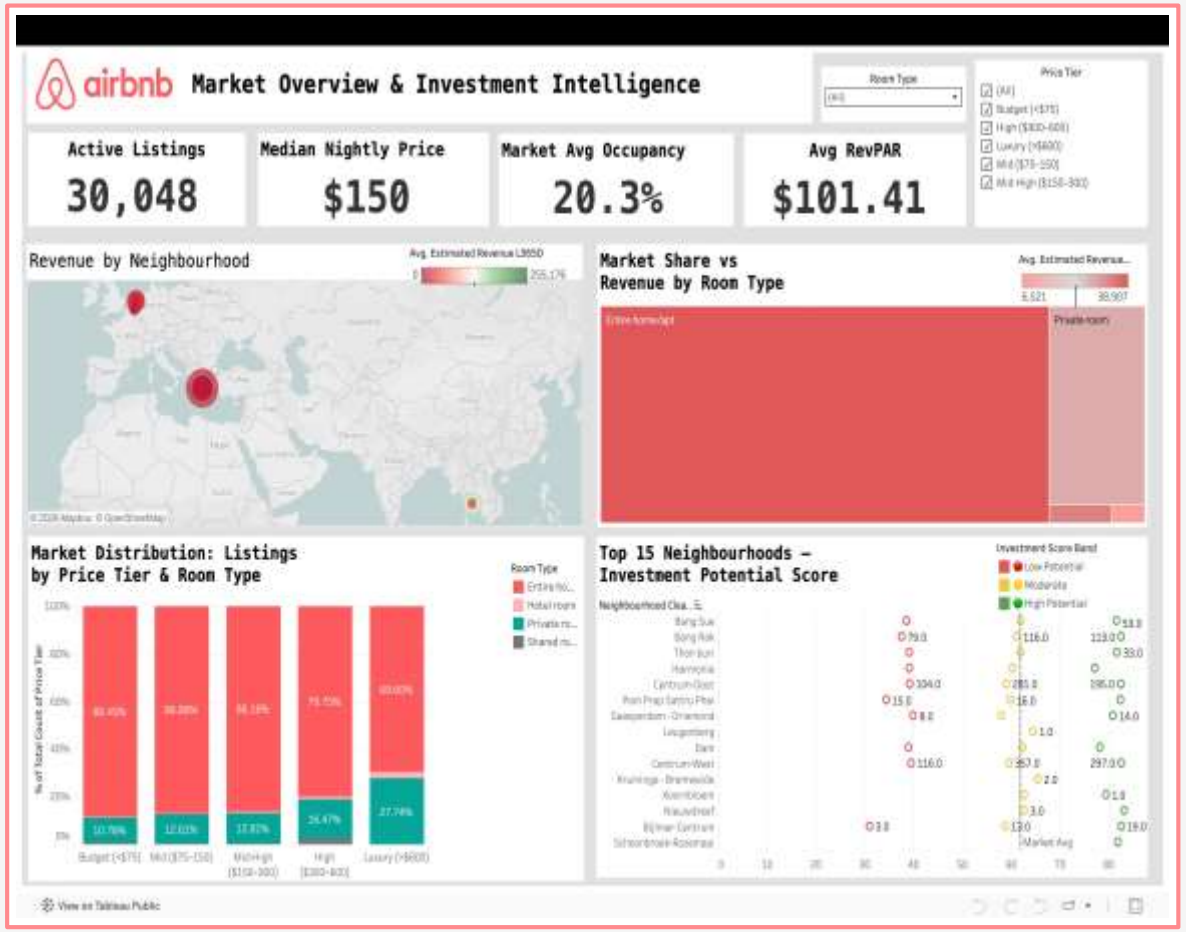


Review Sub-Score Decomposition



DASHBOARD WALKTHROUGH

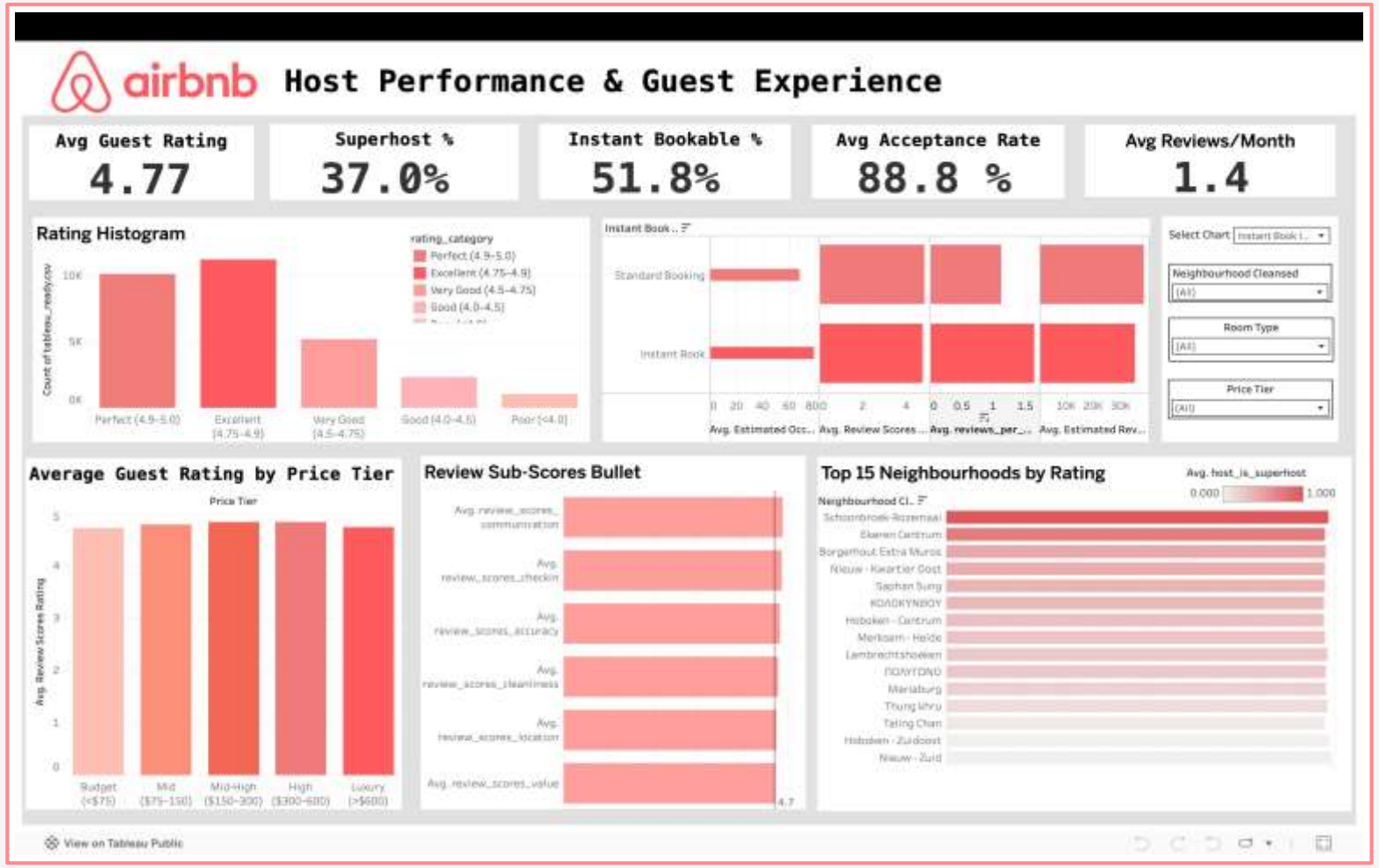
Link



DASHBOARD 1 — Market Overview & Investment Intelligence

DASHBOARD 2 — Revenue & Pricing Strategy

DASHBOARD WALKTHROUGH



WHAT IT ANSWERS

"What makes a 5-star host?"

KEY KPIS

- Avg Rating: 4.77 ★
- Superhost share: 37.0%
- Reviews/month: 1.4
- Instant Bookable: 51.8%
- Acceptance Rate: 88.8%

CHARTS

- Rating histogram by category
- Instant Book vs standard
- Rating by price tier
- Six review sub-scores
- Top 15 neighbourhoods by rating

DASHBOARD 3 — Host Performance & Guest Experience

RECOMMENDATIONS

01.

Become a Superhost

→ Targets the +143% revenue lift
→ Highest-impact controllable action

02.

Fix the 27.1% Zero-Revenue Listings

→ Use mid-tier benchmarks ($75–$300)
→ Re-list inactive properties

03.

Improve the "Value" Sub-Score

→ Currently weakest at 4.705/5.0
→ Biggest competitive differentiator

04.

Target Bangkok Premium Districts

→ Parthum Wan / Bang Rak / Vadhana
→ Best absolute revenue per listing

IMPACT & VALUE

Investor ROI

Investment Potential Score (0–100) standardises 194 neighbourhoods across 4 cities.

Host Revenue Lift

Becoming a Superhost more than doubles revenue: $24K → $58K per year.

Platform Health

Identifying the 8,158 zero-revenue listings exposes a systemic pricing-guidance gap.

Data-Driven Decisions

Three Tableau dashboards replace gut-feel with filterable, visual market intelligence.

IMPACT & VALUE

Impact Snapshot

+143% revenue lift via Superhost
30,048 listings, 194 neighbourhoods covered
$200K+ annual revenue gap addressable
27.1% mispriced listings recoverable

LIMITATIONS & NEXT STEPS

⚠ LIMITATIONS

Scrape-only data — Inside Airbnb is a single February 2026 snapshot, not a continuous feed.

Estimated revenue — derived from price × occupancy proxies, not actual booking ledgers.

No multi-year trend — cannot measure how prices or demand evolve over time.

Currency normalisation — listings denominated in local currencies are simplified to USD via rough conversion.

Unstructured fields dropped — descriptions, reviews and amenities not yet mined.

🚀 FUTURE ROADMAP

ML Price Optimiser — Random Forest / XGBoost model to recommend nightly prices.

Sentiment Analysis — NLP on review text to surface noise, parking and cleanliness pain points.

Seasonal Forecasting — time-series for peak/off-peak demand prediction.

Geographic Expansion — extend to New York, London, Tokyo, Dubai for global comparison.

ROI Calculator Tool — interactive: input purchase + reno costs, output projected annual ROI.

TEAM & CONTRIBUTION

Team Member	Role	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Manan Gilhotra	Project Lead	Y	Y			Y	Y	
Mayank Sharma	Strategy Lead				Y	Y		
Daniel Tayal	ETL Lead	Y	Y			Y		
Vidhi Singhal	Analysis Lead			Y	Y			
Abhiraj Tripathi	PPT & Quality Lead					Y		Y
Khushi Batra	Data Lead			Y			Y	
Mayank Gupta	Visualization Lead				Y	Y		

THANK YOU